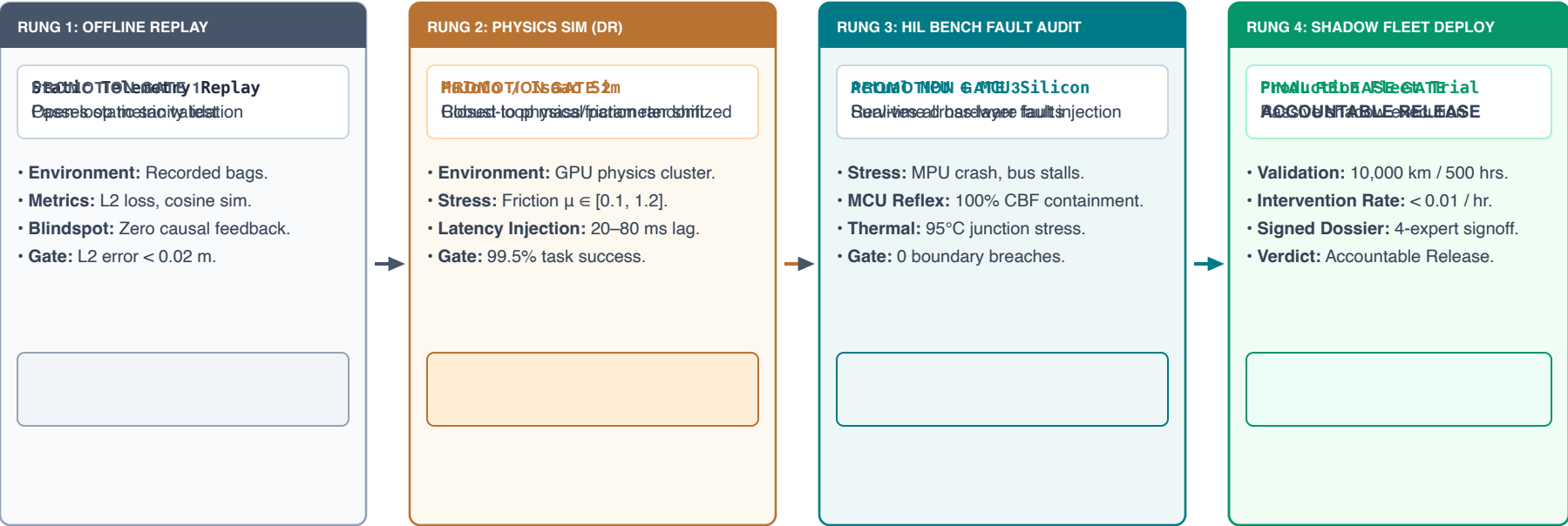


THE 4-RUNG QUALIFICATION LADDER FOR PHYSICAL AI SYSTEMS

Progressive promotion gates: Offline Replay → Physics Sim → Hardware-in-the-Loop Fault Injection → Shadow Fleet Deployment



DEFENSIBLE RELEASE PRINCIPLE (DEMOS ARE NOT PROOF OF SAFETY)

1. **No Skipping Rungs:** A foundation model cannot be deployed to real hardware solely on the basis of leaderboard benchmarks or simulation demos.
2. **Empirical Falsification:** Each promotion gate is designed specifically to expose unmodeled physical, temporal, or silicon failure modes.
3. **Accountable Sign-Off:** Complete release is backed by immutable hardware traces and a Claim-Argument-Evidence (CAE) safety case.